

TechNova Data Analytics Internship

Task 4: Time Series Trend Analysis

1. Introduction

This project analyzes historical AAPL stock-price data using time-series analysis techniques. The objective is to understand price trends, volatility, possible seasonal patterns, unusual movements, and moving-average behavior. Prophet and ARIMA models were also applied as bonus forecasting techniques.

2. Dataset Overview

The dataset contains **1,258 historical AAPL stock records** with information including opening price, closing price, high and low prices, trading volume, adjusted prices, dividends, and split factors.

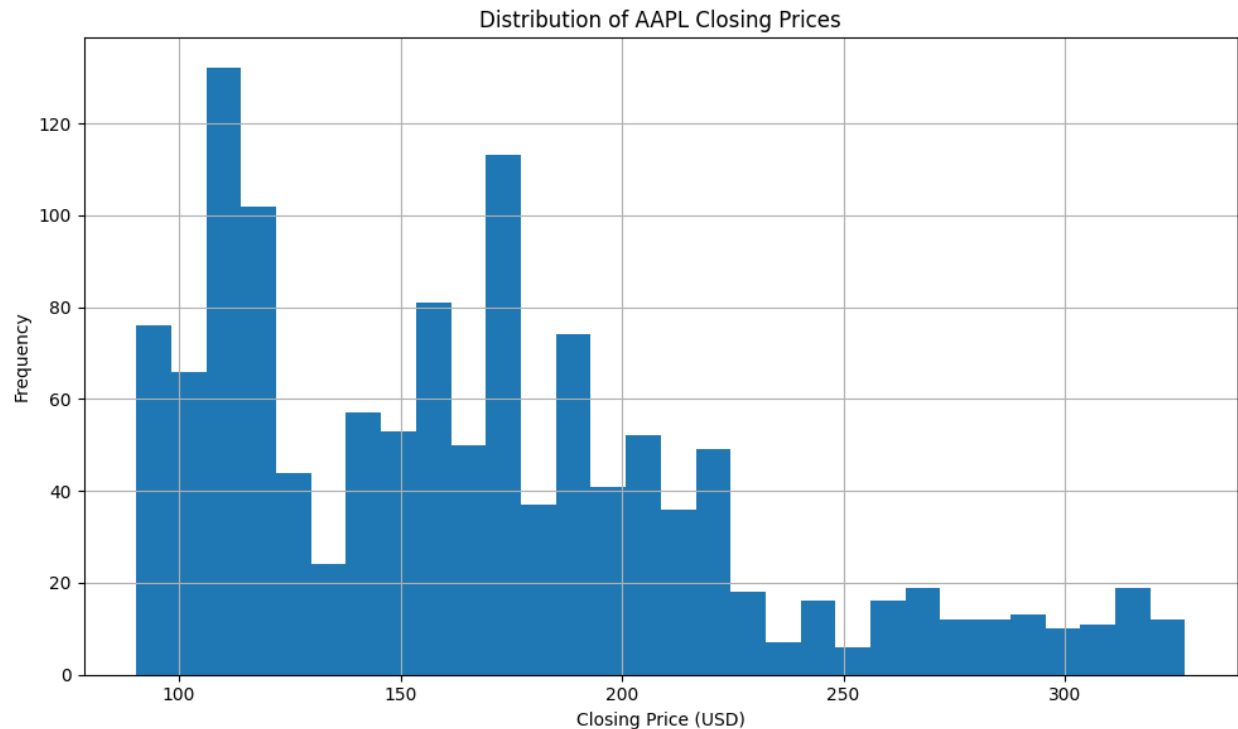
- **Records:** 1,258
- **Stock:** AAPL
- **Missing values:** None
- **Duplicate records:** None
- **Target variable:** Closing price
- **Time-based variable:** Date

The dataset was cleaned by removing the unnecessary index column, converting the date column into datetime format, sorting records chronologically, and preparing the data for time-series analysis.

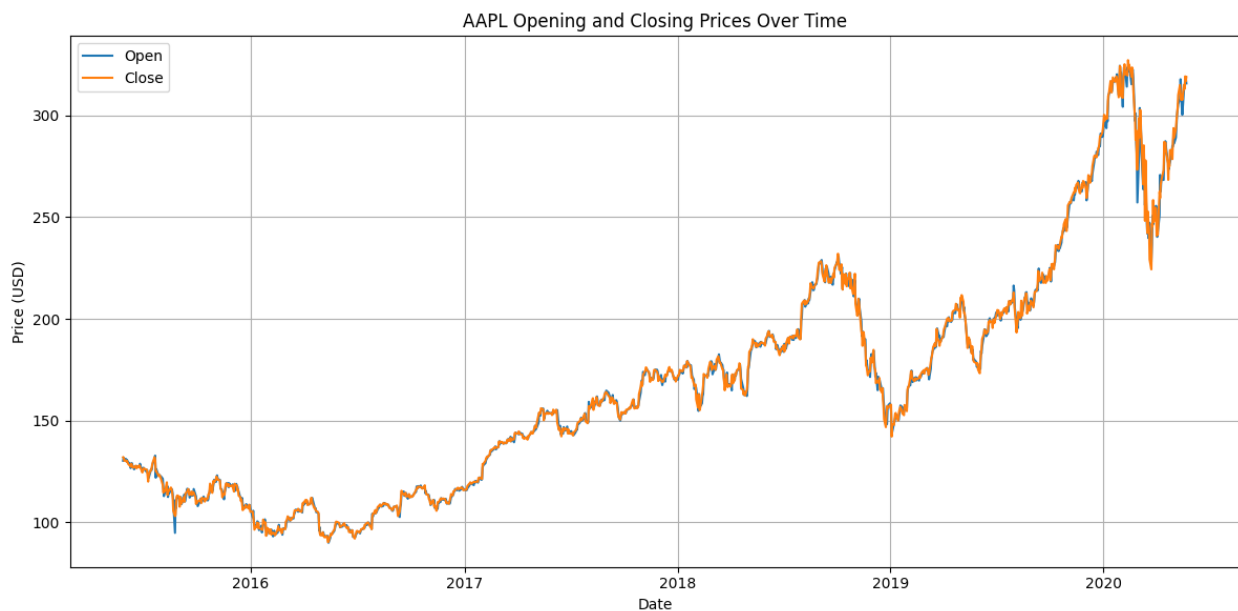
3. Exploratory Data Analysis

Several exploratory analyses were performed to understand the dataset.

The closing price ranged from **\$90.34** to **\$327.20** during the analyzed period. The lowest closing price occurred on **May 12, 2016**, while the highest occurred on **February 12, 2020**.



Opening and closing prices showed similar movements, while differences between them reflected daily price changes. Trading volume also varied considerably, with higher values indicating periods of increased market activity.

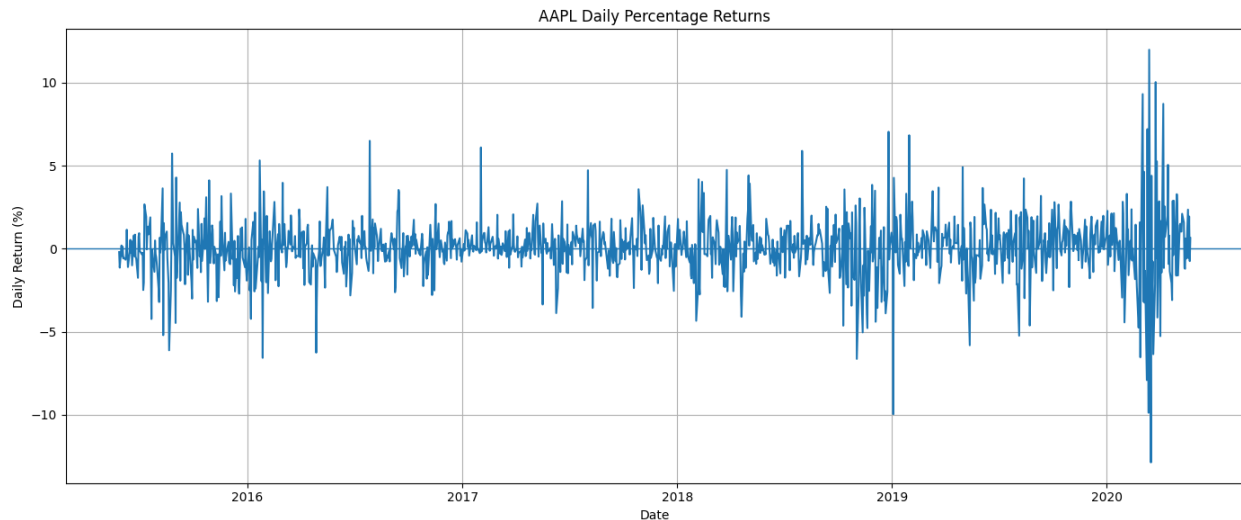


A daily price range was calculated using the difference between the high and low prices to examine short-term volatility.

4. Trend Analysis

The closing-price trend was plotted against time to identify the overall movement of the stock.

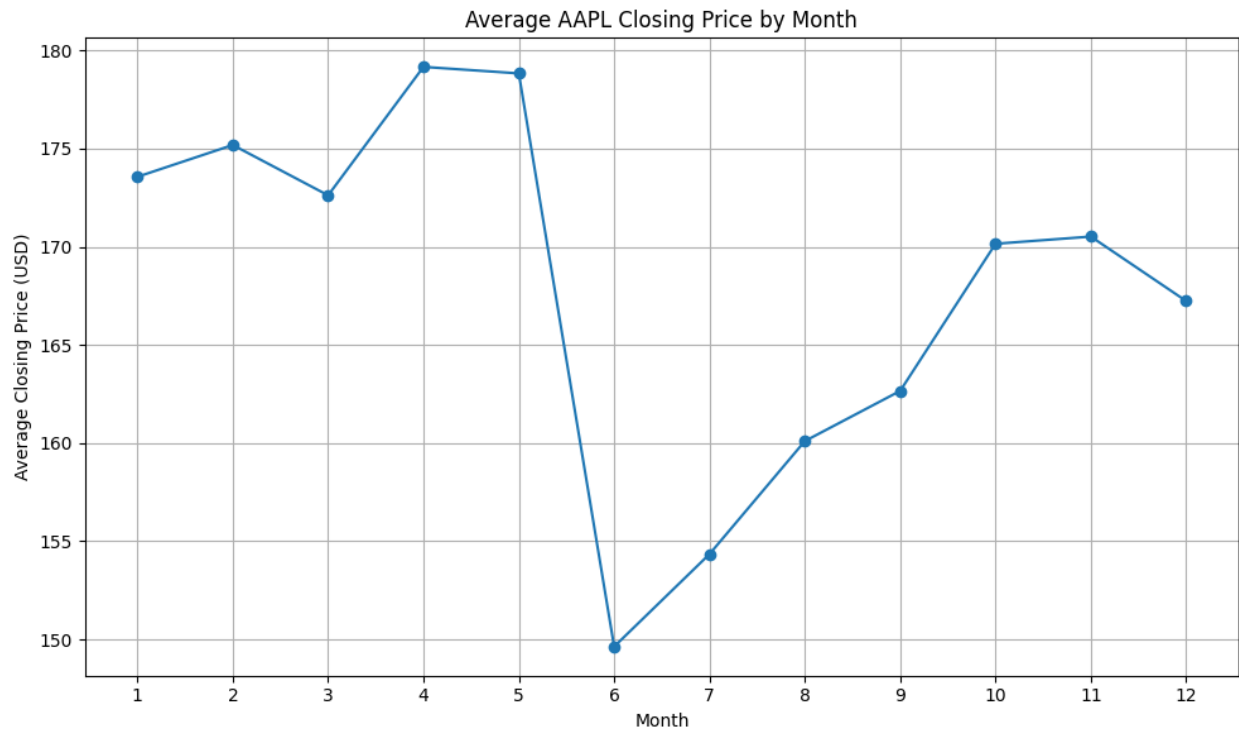
The analysis showed an **overall upward trend with noticeable short-term fluctuations**. Daily percentage returns were also calculated to examine short-term changes. Most daily returns were relatively small, while some trading days showed larger positive or negative movements.



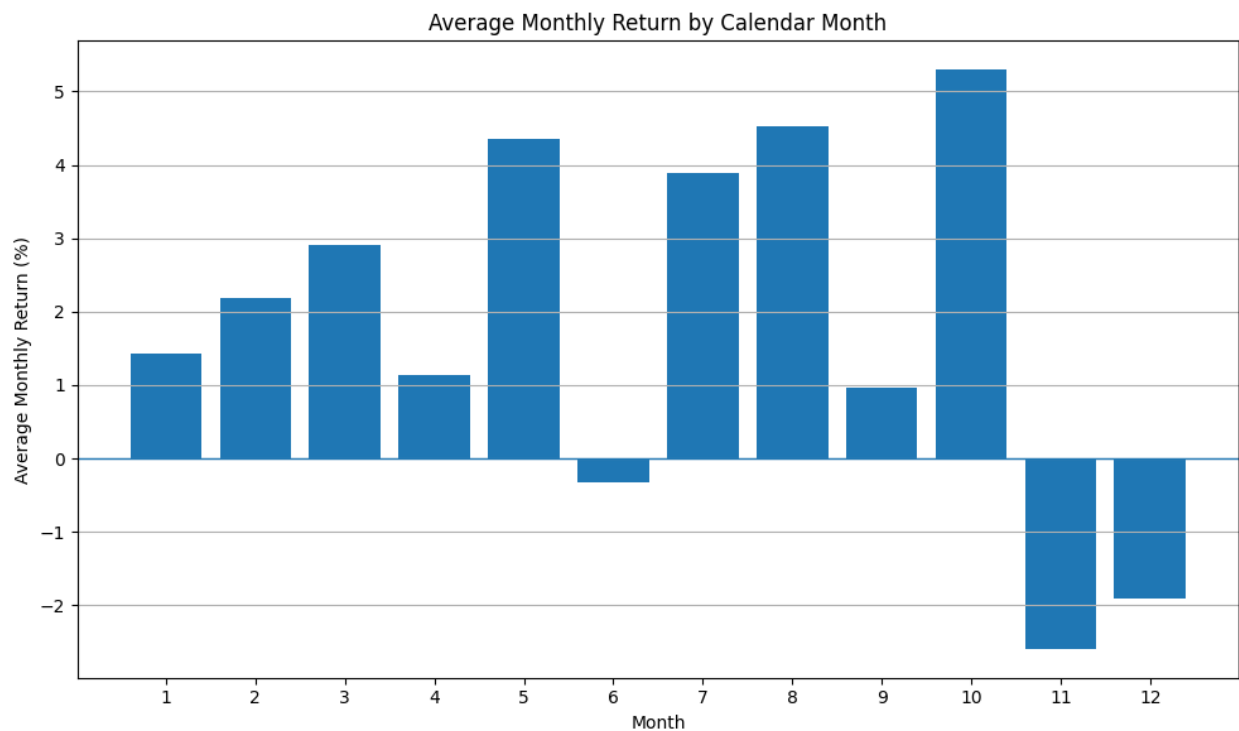
Yearly average closing prices were also examined to provide a broader view of changes in the stock's price level.

5. Seasonality Analysis

Monthly closing prices and monthly returns were analyzed to investigate possible recurring seasonal behavior.

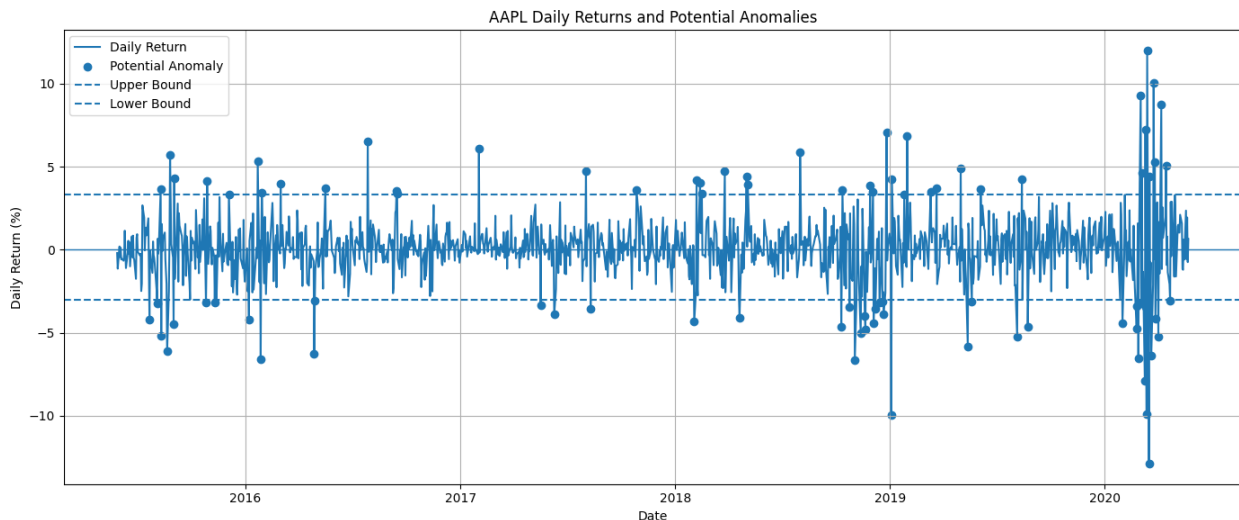


Monthly returns were used because they provide a better indication of recurring patterns than raw monthly prices, which can be affected by the stock's long-term growth. The monthly analysis showed variation between months, but the results do not provide strong evidence of a consistent seasonal pattern.



6. Anomaly Detection

Anomalies were identified using the **Interquartile Range (IQR)** method applied to daily returns.



This method identified trading days where returns were unusually high or low compared with the normal range of daily movements. The largest positive and negative daily returns were also extracted to identify the most significant short-term price movements.

These observations can help highlight periods of unusual market activity for further investigation.

7. Moving Average Analysis

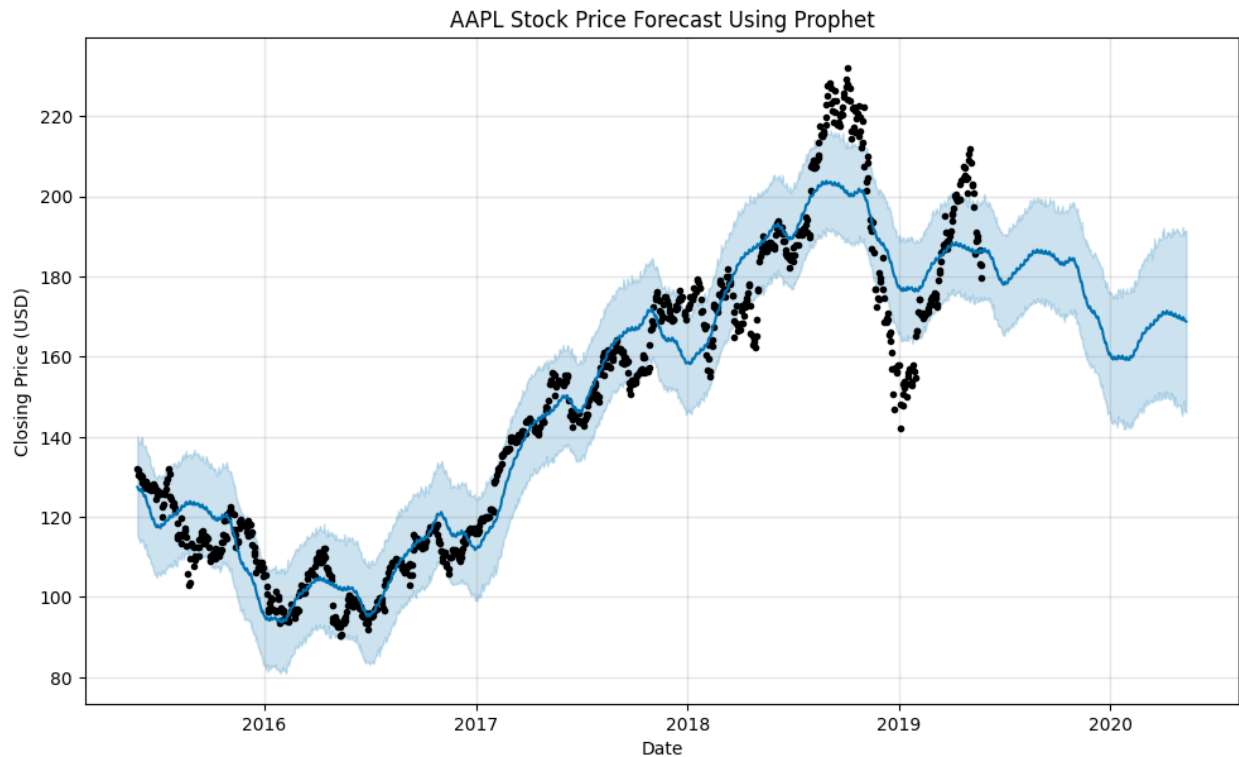
Two moving averages were calculated:

- **7-day moving average:** Captures short-term price movement.
- **30-day moving average:** Provides a smoother view of the broader trend.

The 7-day moving average responds more quickly to recent price changes, while the 30-day moving average reduces short-term fluctuations and provides a clearer view of the underlying trend.

8. Forecasting with Prophet

As a bonus analysis, the dataset was divided into training and testing portions. **Prophet** was trained using historical closing prices and used to forecast the test period.



The forecast was compared with the actual closing prices using:

- Mean Absolute Error (MAE)
- Root Mean Squared Error (RMSE)
- Mean Absolute Percentage Error (MAPE)

This provided a quantitative assessment of the model's forecasting performance.

9. Forecasting with ARIMA

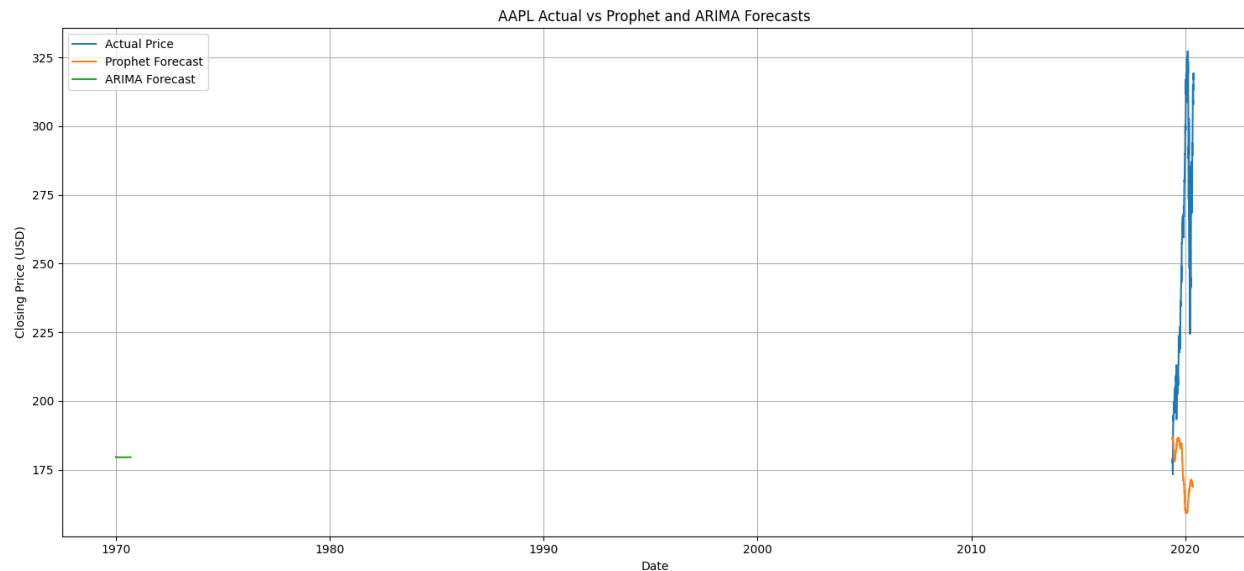
An **ARIMA (5,1,0)** model was also applied to the closing-price time series. The model generated forecasts for the same test period used for Prophet.

ARIMA predictions were evaluated using MAE, RMSE, and MAPE, allowing the forecasting performance of the two models to be compared on the same test data.

10. Model Evaluation

Prophet and ARIMA forecasts were compared using the calculated error metrics.

The model with lower error values provides better performance **for this particular test period and dataset**. The comparison demonstrates how different time-series forecasting approaches can be evaluated objectively rather than relying only on visual inspection.



11. Key Findings

- AAPL showed an overall upward price trend during the analyzed period.
- The closing price varied substantially, ranging from **\$90.34 to \$327.20**.
- Daily returns revealed periods of higher short-term volatility.
- IQR analysis identified unusually large price movements.
- The 7-day moving average captured short-term changes more quickly.
- The 30-day moving average provided a smoother representation of the broader trend.
- Monthly analysis showed variation between months without strong evidence of consistent seasonality.
- Prophet and ARIMA provided two different approaches for forecasting future prices.

12. Business Insights

- Trend analysis can help monitor changes in overall market price levels.
- Moving averages can support monitoring of short- and medium-term trends.
- Anomaly detection can highlight unusual periods of market activity.
- Trading-volume changes can provide additional context for significant price movements.
- Forecasting models can support data-driven scenario planning.
- Combining trend, volatility, volume, and forecasting analysis provides a more complete view of stock-price behavior.

13. Recommendations

- Continue monitoring moving averages to identify changes in price trends.
- Investigate unusually large returns and volume spikes for potential market events.
- Update forecasting models as new stock-price data becomes available.
- Compare multiple forecasting models rather than relying on a single approach.
- Evaluate model performance regularly using MAE, RMSE, and MAPE.

14. Conclusion

This project successfully performed time-series analysis on AAPL stock data by examining trends, volatility, seasonality, anomalies, and moving averages. Prophet and ARIMA were additionally used for forecasting and evaluated using standard error metrics. Overall, the analysis demonstrates how Python-based time-series techniques can be used to understand historical stock behavior and support data-driven analysis.